1. What is white noise?

Definition: White noise is a type of time series data where it has a constant mean, constant variance, and for any time periods, u_s is uncorrelated with u_j where $j \neq s$

Therefore, white noise must be a stationary series.

(b) and (c)

Sketch of Proof:

If we take the expectation on both sides of $y_t = a_0 + a_1 y_{t-1} + u_t$, then.

$E(y_t) = E(a_0 + a_1 y_{t-1} + u_t)$

$= a_0 + a_1 E(y_{t-1}) = a_0 + a_1 E(a_0 + a_1 y_{t-2} + u_{t-1})$

$=a_0 + a_1 a_0 + a_1^2 E(y_{t-2}) = a_0 + a_1 a_0 + a_1^2 E(a_0 + a_1 y_{t-3} + u_{t-3})$

$=a_0 + a_0 a_1 + a_0 a_1^2 + a_1^3 E(y_{t-3})$

Repeating the steps to sub $y_{t-3}, \ldots . y_{t-n}$

$=a_0(1+a_1 + \cdots a_1^{n-1}) + a_1^n E(y_{t-n})$

When the series is long enough, for example $n \to \infty$, and $|a_1| < 1$, then $a_1^\infty = 0$,

$E(y_t) = a_0(1+a_1 + \cdots a_1^{n-1} + \cdots) + 0$

The sum of this infinite geometric series can be calculated easily.

$$E(y_t) = a_0/(1 - a_1)$$

Similarly, we can take the variance on both sides for this $AR(1)$ model.

$$Var(y_t) = Var(a_0 + a_1 y_{t-1} + u_t)$$

$$Var(y_t) = 0 + a_1^2 Var(y_{t-1}) + Var(u_t)$$

$$\Rightarrow (1 - a_1^2) Var(y_t) = \sigma^2$$

$$\Rightarrow Var(y_t) = \frac{\sigma^2}{1 - a_1^2}$$

2.

The last five lecture notes covered (A) Cross-sectional Models and (B) Time Series Models

Why cross-sectional models?

(i) Data availability. We only have cross-sectional units.

(ii) Want to find out ceteris-paribus relationships and to conduct hypothesis test

Therefore, understanding the 6 assumptions (LN1, Sections C, D and LN2 Section A) is important.

Why time series models?

(i) Data availability/ Nature of the topics

(ii) Want to find out the impact of X on Y (and hypothesis testing is involved here)

(iii) Want to do forecasting

Therefore, understanding the 6 assumptions (LN3 Section B) and if they are satisfied are important.

Stationarity is crucial concept!

What time series models did we cover?

(a) Static Model: $y_t = \beta_0 + \beta_1 x_{t1} + \beta_2 x_{t2} + u_t$

(b) FDL Model: $y_t = \alpha_0 + \delta_0 x_t + \delta_1 x_{t-1} + \delta_2 x_{t-2} + \cdots + \delta_p x_{t-p} + u_t$

(c) AR(p) Model: $y_t = a_0 + a_1 y_{t-1} + a_2 y_{t-2} + \cdots + a_p y_{t-p} + u_t$

We can also combine (b) and (c) models. For example,

$$y_t = a_0 + a_1 y_{t-1} + a_2 y_{t-2} + \delta_0 x_t + \delta_1 x_{t-1} + u_t$$

Which one should be checked first?

(I) Stationarity? Or No Serial Correlation and Homoskedasticity?

(II) If u_t is a white noise, then Assumption TS 4 and 5 hold.